

Matrix Representation of Edges

It is often convenient to represent edges by a matrix called **adjacency matrix**.

The adjacency matrix for $G = \langle V, E \rangle$ is the following $n \times n$ matrix: $a_{ij} = 1$ if $(i, j) \in E$ and $a_{ij} = 0$ when $(i, j) \notin E$. We assume that there is no self-link, $a_{ii} = 0$ for any i (equivalently $(i, i) \notin E$), unless noted otherwise.

G is an undirected graph $\Leftrightarrow$ its adjacency matrix A is **symmetric** (i.e. $a_{ij} = a_{ji}$).

Example: Adjacency matrix for

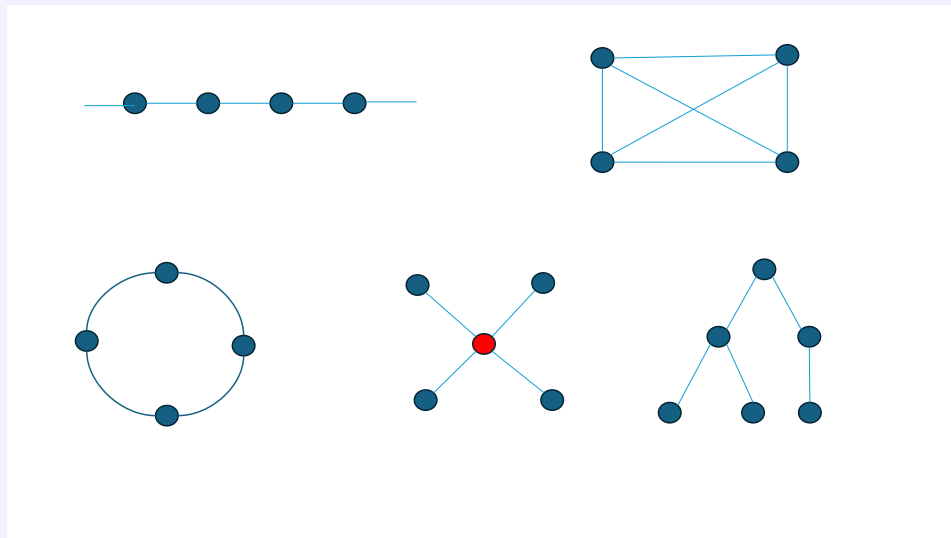
$$V = \{1, 2, 3\}, E = \{(1, 2), (1, 3), (2, 3), (3, 1), (3, 2)\}$$

$$\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Special Networks

Here are some special graphs/networks. All undirected.

- **Line**
- **Cycle**
- **k -Regular graph:** Every node is linked to k other nodes.
- **Clique (Complete network):** Every node is linked to every other node.
- **Star:** One center is linked to all the other nodes, which are linked only to the center.
- **Tree and Forest:** Tree is a **connected** graph without any **cycle**. Forest is a graph where every **component** is a tree.



Which is which?

Path and Connectivity

- A **path (walk)** from node i to j is a sequence of edges $(i_1, i_2), (i_2, i_3), \dots, (i_{K-1}, i_K)$ where $i_1 = i$ and $i_K = j$.
- **Simple path (path)** is a path along which the same node does not appear twice.
- **Cycle** is a (nonsimple) path where the same node appears only once at the beginning and the end of the path.

Among the previous examples of networks, which one has the unique path between every pair of nodes? Which network has or does not have a cycle?

Path and Connectivity

- i and j are **connected** if there is a path from i to j or a path from j to i .
- Graph G is **connected** if every pair of nodes is connected. For directed graph G specifically, we say that G is **strongly connected** if there is a (directed) path from any node to any other node.
- $G' = \langle V', E' \rangle$ is a **subgraph** of $G = \langle V, E \rangle$ if $V' \subset V$ and E' is a collection of edges from E that involves nodes V' .
- **(Connected) Component** is a (nonempty) connected subgraph that is not a part of any larger connected subgraph.
- **Giant component** is a large connected component that contains a significant fraction of all nodes.

Measuring Networks

The space of networks is so huge. If each one of them is very different from any other, then there would be no hope for us to understand networks systematically.

It would be useful to have some simple measurements/parametrization of networks to reduce the dimension of the space of networks.

We introduce some quantitative measurements of networks. They are at least useful as summary statistics to classify data, and hopefully capture some local/global properties of networks.

Distance

We are often interested not only whether certain nodes are connected or not, but also how “close” they are. For example, the idea of “small-world phenomenon” is that you can reach anyone in the vast social network through a surprisingly short path.

We like to introduce a few formal definitions and make the notions of distance precise.

- The **length** of a path is the number of edges along the path.
- The **distance** between two distinct nodes is the length of the shortest (simple) path between the nodes. Denote the distance between $i, j \in V$ by $d(i, j)$.

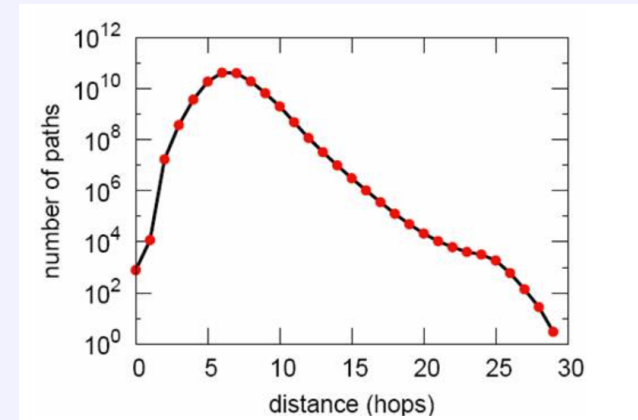
For a whole (**connected**) network, we define the following summary statistic regarding distances.

- **Diameter** of G is the largest distance between two nodes in G . Formally, it is $\max_{(i,j) \in E} d(i, j)$.
- **Average distance** of G is the average of all the distances across all the pair of nodes, i.e. $\frac{\sum_i \sum_{j \neq i} d(i, j)}{n(n-1)}$.

What is the connected network with n nodes with the largest/smallest diameter/average distance?

Six-Degree of Separation Revisited

- Milgram's experiment provides a support for a small-world hypothesis. But it is far from conclusive due to the limited data and possible bias.
- Leskovek and Horvitz (2008) studies a huge (undirected) network of 240 millions Microsoft Instant Messenger users (likely captures the half of all conversations!).
- Among other things, they found that....
 - ▶ 99.9% of the users belong to the giant component.
 - ▶ the average distance of the network is 6.6 and its median distance is 6.
 - ▶ there is a significant homophily (more conversation with similar people), stronger in the order of Language, Geographic location, Age. Gender is an exception; "negative" homophily for gender.



The distribution of distances of IM users, Leskovec and Horvitz (2007).

Connectivity and Distance via Adjacency Matrix

Connectivity and distance of a network can be captured by its adjacency matrix.

- We consider directed graphs. Remember that $a_{ij} = 1$ for adjacency matrix A represents a link/arrow from i to j .
- Who can i reach by a path of length 1? They are j such that $a_{ij} = 1$.
- Then who can i reach by a path of length 2? They are j such that $a_{ik} = a_{kj} = 1$ for some $k \neq i, j$. Equivalently, they are j such that $\sum_k a_{ik} a_{kj} > 0$. Note that some node can be reached by a path of length 1 and a path of length 2.
- Note that $\sum_k a_{ik} a_{kj} = 1$ is ij -element of $n \times n$ matrix A^2 , which we denote by $a_{ij}(2)$. More generally, $a_{ij}(t)$ counts the number of paths of length t from i to j .

For the previous example, A^2 is the following 3×3 matrix.

$$A^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

- If $a_{ij}(2) = 0$, then there is no path of length 2 from i to j .
- If $a_{ij}(2) > 0$, then there is a path of length 2 from i to j .
- If $a_{ij}(2) = m$, then there are m distinct paths of length 2 from i to j .

More generally, observe the following.

- There is a path from i to j if and only if there is some t such that $a_{ij}(t) > 0$.
The smallest such t is the distance between i and j .
- Graph G is strongly connected if and only if **there is some t such that every nondiagonal element of $\sum_{s=1}^t A^s$ is positive**. The smallest such t is the **diameter of G** .
- **One special case is that there exists t such that A^t is a positive matrix (i.e. every element of A^t is a positive number). This means that every node can be reached from every node exactly in t steps.** (If A^t is a positive matrix, then $A^{t+1}, A^{t+2}, \dots$ are all positive matrices. Do you see why?)

Degree

It is natural to think that more connected people are more influential and play an important role in the community.

Then we may be interested in the distribution of such connections in the community.

Degree is the simplest measure of such connectedness.

Neighbor and Degree

- i 's **neighbor** N_i is a subset of nodes that are linked to i . Formally
 $N_i = \{j \in V \mid (i, j) \in E \text{ or } (j, i) \in E\}$.
- **Degree** d_i of i is the size of i 's neighborhood, i.e. $d_i = \#N_i$.

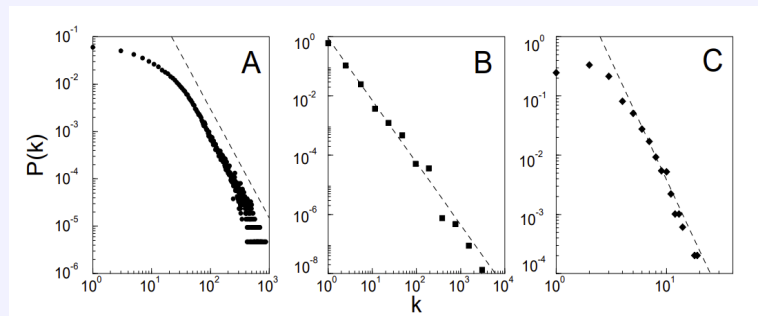
For digraph, we distinguish between **in-degree** and **out-degree**.

- **in-degree:** The in-degree d_i^- of i is the number of links toward i . Formally,
 $d_i^- = \#\{j \in V \mid (j, i) \in E\}$ or $d_i^- = \sum_{j \neq i} a_{ji}$.
- **out-degree:** The out-degree d_i^+ of i is the number of links from i .
Formally, $d_i^+ = \#\{j \in V \mid (i, j) \in E\}$ or $d_i^+ = \sum_{j \neq i} a_{ij}$.

Degree Distribution

- Let n_k be the number of nodes that have k neighbors.
 $P(k) = \frac{n_k}{n}$, $k = 0, \dots, n-1$ is the **degree distribution** of network.
- The degree distribution of real social networks is usually fat-tailed, driven by many well-connected individuals at the tail.
- They can be approximated by **power-law** distributions: $P(k) = ck^{-\gamma}$, where c is the scaling factor.

Power law distribution has a nice property: it becomes linear in $\log k$:
 $\log P(k) = \log c - \gamma \log k$. γ becomes its slope, so can be easily estimated.



The degree distributions of large networks: A: Actor collaboration graph, B: WWW, C: Power grid, Barabási and Albert (1999)

Remark: some properties of power-law distributions

- It is **scale-free**. Observe $\frac{P(2k)}{P(k)} = \frac{P(4k)}{P(2k)}$. Whenever a degree is twice as large, the probability decreases by the same percentage.
- The mean degree exists only when $\gamma > 2$. Estimated γ is usually above 2
- Note that $P(k+1)/P(k) = \left(\frac{k}{k+1}\right)^\gamma$. So, the tail probability is decreasing slowly.
 - To put this into perspective, consider the following random benchmark. Imagine that a node can be linked with each of $n-1$ other nodes with probability p independently. Set $p = \frac{\lambda}{n}$ so that the expected degree matches with the average degree λ . Then the distribution of degrees becomes approximately Poisson distribution ($P(k) = \frac{\lambda^k e^{-\lambda}}{k!}$) when n is large. For this case, $P(k+1)/P(k) = \frac{\lambda}{k+1}$.

Clustering Coefficient

Many researchers have found that two individuals who are linked to the same individual tend to be linked more than a random pair of individuals. That is, if we know that A and B has a friend in common, then we can expect that A and B are friends.

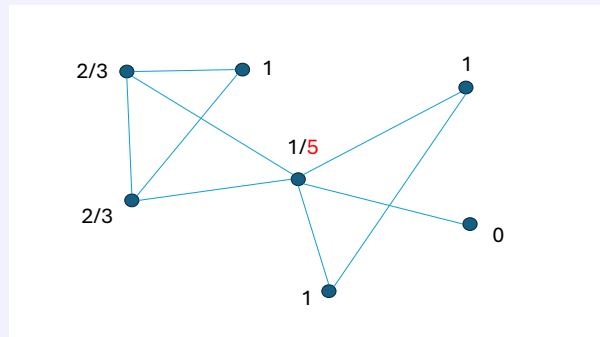
We introduce a formal metric of the prevalence of such “triads” and clarify what “tend to be” means.

Clustering Coefficient

- In your friendship network, your **clustering coefficient** is the ratio

$$\frac{\text{the number of all pairs of your friends who are actually friends}}{\text{the number of all possible pairs of your friends}}.$$
- Formally, clustering coefficient of i is $C_i = \frac{\sum_{j \neq k \neq i} a_{ij} a_{ik} a_{jk}}{\sum_{j \neq k \neq i} a_{ij} a_{ik}} = \frac{\sum_{j \neq k \neq i} a_{ij} a_{ik} a_{jk}}{d_i(d_i - 1)}$
- There are two ways to define the average clustering coefficient.
 - One is simply $\frac{\sum_{i=1}^n C_i}{n}$.
 - The other one is $\frac{\sum_i \sum_{j \neq k \neq i} a_{ij} a_{ik} a_{jk}}{\sum_i \sum_{j \neq k \neq i} a_{ij} a_{ik}}$ (What is this?)

Clustering Coefficient: Example



- It is known that clustering coefficient is relatively high in co-authorship networks (different across fields, Physics > Econ > Math).
- Goyal, van der Leij, and Moraga-González (2006) reports that the coefficient is roughly 0.2 for Economists. Again if everything is completely random, we expect that a link between two nodes is formed with probability $p = 1/20000$ given roughly 34000 nodes with the average degree 1.7. This is much smaller than the observed clustering coefficient 0.2.

Each number is an individual clustering coefficient. What is the average clustering coefficient?

Summary

- Basic terminology of graph theory
- Path and connectedness
- Various metrics about networks: 1. Distance, 2. Degree, 3. Clustering Coefficient

Suggested reading. EK Ch.2